Theory of Open Systems

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Chapters:

1. Density operators: pure states, mixed states, ensembles, purity
2. BlochSphere: Geometric representation of Density Matrix
3. Decoherence:
4. Kraus operators & Quantum channels: phase damping bit flip, depolarization.
5. Lindblad master equation: derivation, connection to Kraus operators

Chapter 1: Density Matrix

Why we need Density Matrix? Why not just EigenVectors?

Pure State

- if our ensemble is well known quantum state
  i.e $|\psi\rangle = \sum_i c_i |\lambda_i\rangle$
- then when we measure will find probability of getting eigen value λ_i is $|C_i|^2$
- $\langle \lambda_i \rangle = |C_i|^2$
- Note: Only Eigenvector representain. can explain QM. because this representation includes property of Supperposition.

Mixed State

- if our ensemble is a mixture of many pure states $\{|\psi^k\rangle, q_k\}$ and each $|\psi^k\rangle = \sum C_i^k |\lambda_i\rangle$
- then when we measure we dont know which state of $\{|\psi_k\rangle\}$ gave us the eigen value λ_i
- We could only get an average $\boxed{\langle \lambda_i \rangle = \sum_k \lambda_i q_k |C_i^k|^2}$

Side Tack but Useful construction: Mathematical Description of Measurement ?

According To David Tong , in his book. Describes The act of measurement is One of the more mysterious aspect of quantum mechanics .

- Measurement cannot be represented as an Unitary Evolution.
- It abruptly collapse the wavefuntion

Projective Measurements

Any Observable $\hat{O}$ can be decomposed as $\hat{O} = \sum_m \lambda_m |m> < m|$,

This is called Spectral Theorem / Spectral Representation.

Where $|m>$ is the basis off the Eigen-Vector-space V which the operator acts on

And λ_m is the eigenvalue of Operator $\hat{O}$.

We can define $|m><m| := \hat{M}$ thus $\hat{O} = \sum_m \lambda_m \hat{M}$

Note: $\hat{M}_n \hat{M}_m = \hat{M}_n \delta_{mn}$ and $\sum_m M_m = I$. Since $\{|m>\}$ space the Eigenvector-space

- Axiom 1: Born Rule : probability of getting λ_m is $p(k) = <\psi|M_k|\psi>$
- Axiom 2: The act of measurement has disturbed the state

$$|\psi> \longrightarrow |\psi_k> = \frac{M_k|\psi>}{\sqrt{p(k)}}$$

Now coming back to Density Operator:

Probability of getting eigenvalue $\lambda_k = \sum_i q_i < \psi_i | M_k^\dagger M_k | \psi_i > = Tr[M_k^\dagger M_k \rho]$

Where $\rho = \sum_i q_i | \psi_i > < \psi_i |$ which is known as Density Matrix.

Since we know a measurement operator M_k results $| \psi > \longrightarrow | \psi_k > = \dfrac{M_k | \psi >}{\sqrt{p(k)}}$

Which suggest $\rho \longrightarrow \rho_k = \dfrac{M_k \rho M_k^\dagger}{p(k)}$ where $p_k = Tr[\rho M_k^\dagger M_k]$

The Benefit of using ρ (i.e Density Matrix) is that we can represent

both Mixed -State and Pure state in one construction. If ρ pure then $\rho^2 = \rho$

Chapter 2. Bloch Sphere

Geometric Representation of ρ

Any $d \times d$ Density Metric can be Geometrically represented as $d^2 - 1$ dimensional Sphere (Or Deformed Sphere) .

Example :

A 2×2 Density matrix Can be represented by $\vec{v} = (v_x, v_y, v_z) \; \Big| \; |v| \leq 1$

i.e $\quad \rho = \frac{1}{2}(I + \vec{v} \,.\, \vec{\sigma})$

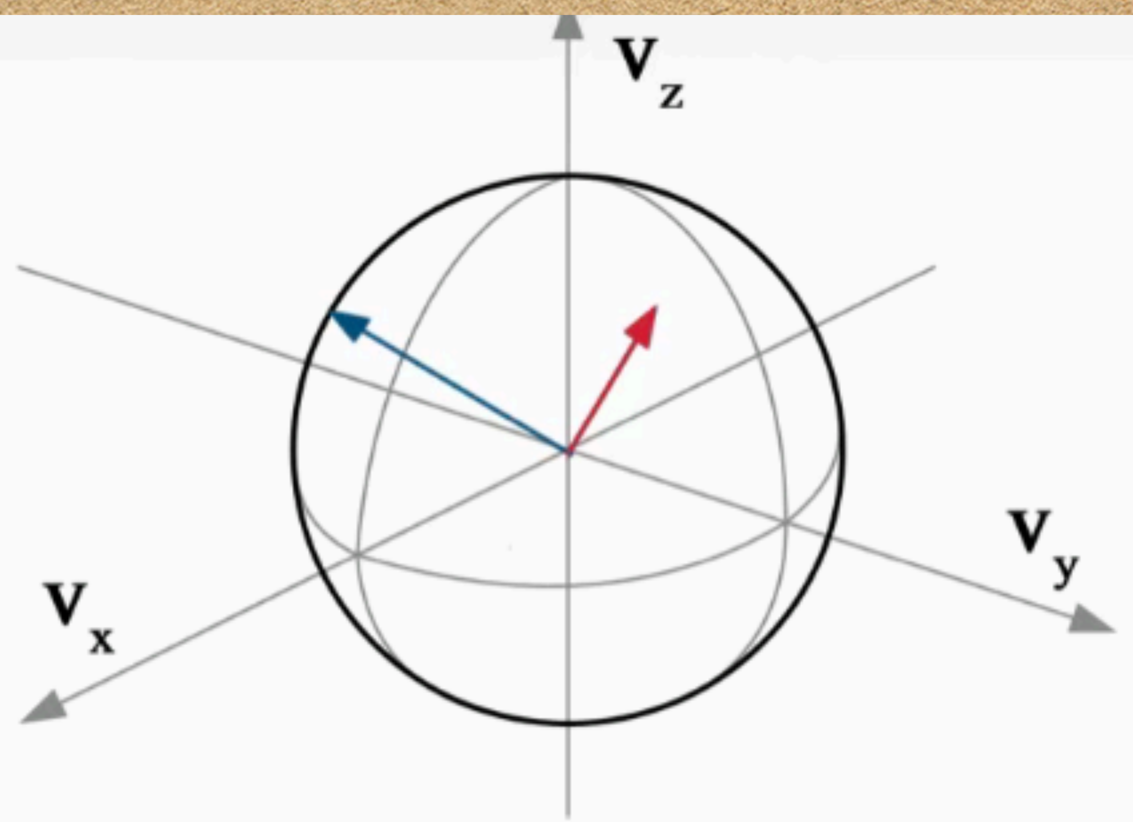


FIG. 1. The Bloch sphere is a geometric representation of the collection of all Bloch vectors $\vec{v}$ which describe valid qubit density operators. Thus, the sphere is of radius 1, its surface represents all pure states, and its interior represents all mixed states. In this diagram the blue vector lies on the surface of the sphere indicating a pure state, whereas the red vector lies in its interior indicating a mixed state.

- Z poles ($\vec{v} = (0, 0, \pm 1)$): The density matrix takes the form

$$\begin{aligned}\rho &= \frac{I \pm \sigma_z}{2} \\ &= \frac{1 \pm 1}{2}|0\rangle\langle 0| + \frac{1 \mp 1}{2}|1\rangle\langle 1|\end{aligned}$$

  which yields $|0\rangle\langle 0|$ for $v_z = 1$ and $|1\rangle\langle 1|$ for $v_z = -1$.

- X poles ($\vec{v} = (\pm 1, 0, 0)$): The density matrix takes the form

$$\begin{aligned}\rho &= \frac{I \pm \sigma_x}{2} \\ &= \frac{1}{2}\left(|0\rangle\langle 0| + |1\rangle\langle 1| \pm \left(|0\rangle\langle 1| + |1\rangle\langle 0|\right)\right) \\ &= \frac{1}{2}\left(|0\rangle \pm |1\rangle\right)\left(\langle 0| \pm \langle 1|\right)\end{aligned}$$

  which yields $|+\rangle\langle +|$ for $v_x = 1$ and $|-\rangle\langle -|$ for $v_x = -1$.

- Y poles ($\vec{v} = (0, \pm 1, 0)$): The density matrix takes the form

$$\begin{aligned}
\rho &= \frac{I \pm \sigma_y}{2} \\
&= \frac{1}{2}\left(|0\rangle\langle 0| + |1\rangle\langle 1| \pm (-i|0\rangle\langle 1| + i|1\rangle\langle 0|)\right) \\
&= \frac{1}{2}\left(|0\rangle \pm i\,|1\rangle\right)\left(\langle 0| \pm (-i)\,\langle 1|\right)
\end{aligned}$$

which yields $(|0\rangle + i\,|1\rangle)/\sqrt{2}$ for $v_y = 1$ and $(|0\rangle - i\,|1\rangle)/\sqrt{2}$ for $v_y = -1$.

- Center ($\vec{v} = (0, 0, 0)$): The density matrix takes the form $\rho = I/2$, the maximally mixed state.

If the Density matrix is pure and $|\vec{v}| = 1$, its $Tr[\rho^2] = 1$.

Chapter 3: Decoherence

The initial System + Bath State is : $|\Psi_i> = \alpha|s_1>|E_0> + \beta|s_2>|E_0>$

The Final Composite State is : $|\Psi(t)> = \alpha|s_1>|E_1> + \beta|s_2>|E_2>$

$$\begin{aligned}\rho_S &= \mathrm{Tr}_E(\rho_{SE}) = \mathrm{Tr}_E|\Psi\rangle\langle\Psi| \\ &= |\alpha|^2\,|s_1\rangle\langle s_1| + |\beta|^2\,|s_2\rangle\langle s_2| \\ &\quad + \alpha\beta^*|s_1\rangle\langle s_2|\langle E_2|E_1\rangle + \alpha^*\beta|s_2\rangle\langle s_1|\langle E_1|E_2\rangle.\end{aligned}$$

$$\rho_S(t) = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* < E_2|E_1> \\ \alpha^*\beta < E_1|E_2> & |\beta|^2 \end{bmatrix}$$

Thus probability density of position x:

$$\begin{aligned} p(x) &= \mathrm{Tr}_S(\rho_S x) \\ &= |\alpha|^2 |\psi_1(x)|^2 + |\beta|^2 |\psi_2(x)|^2 \\ &\quad + 2\,\mathrm{Re}\left\{\alpha\beta^* \psi_1(x)\psi_2^*(x)\langle E_2|E_1\rangle\right\}, \end{aligned}$$

Hence when $< E_1 | E_2 > = 0$ (i.e perfectly distinguishable)

the interference term vanishes is now the system behave like classical particle

when $< E_1 | E_2 > = 1$ (i.e perfectly In-distinguishable)

the interference term at Max and now the system behave like wave

Let Environment has N particles $\implies |E_i> = \Pi_k^N |e_k>$

Thus $<E_1|E_2> = \Pi_k^N \ <e_i^k|e_j^k>$ assuming each $<e_i^k|e_j^k> = \mu \mid \mu < 1$

$\implies <E_1|E_2> = (1-\epsilon)^N = (1-\frac{\Gamma t}{N})^N \approx e^{-\Gamma t}$ for large N

$$\rho_S(0) = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix} \xrightarrow{\Phi} \rho_S(t) = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* e^{-\Gamma t} \\ \alpha^*\beta e^{-\Gamma t} & |\beta|^2 \end{bmatrix}$$

Chapter 4: Kraus Operator & Quantum Maps

Suppose there exist a map $\Phi : \rho(0) \longrightarrow \rho(t)$ (Here we can treat ρ as an operator $\hat{X} \in H_S$)

and Φ follows this following property:

1. Trace Preserving : i.e $Tr[\Phi(\rho(0))] = Tr[\rho(0)]$

2. Linear : i.e $\Phi(a\rho_1 + b\rho_2) = a\Phi(\rho_1) + b\Phi(\rho_2)$

3. Positivity : i.e Φ maps any positive operator to a positive operator

If Φ is Complete Positivity : I.e let there be a k dim Hilbert space of a second system H_R.

Thus new map for the combined system will be $\Phi \otimes I_k$,

if the $\Phi \otimes I_k$ maps positive to positive operators $\forall k$ then its Complete Positive map

Any Φ which is Trace-Preserving, Linear and Complete-Positivity can be represented as $\Phi(X) = \sum_a K_a X K_a^\dagger$ with $\sum_a K_a K_a^\dagger = I$

So one can Express $\rho_s(t) = \sum_a K_a(t)\rho_s(0)K_a^\dagger(t)$, where $K_a := Kraus - Operator$

$$
\begin{tikzcd}
\rho_S(0) \otimes \rho_B(0) \arrow[r, "\mathcal{U}"] & U\left[\rho_S(0) \otimes \rho_B(0)\right]U^\dagger \arrow[d, "\mathrm{Tr}_B"] \\
\rho_S(0) \arrow[u, "\mathcal{A}"] \arrow[r, dashed, "\Phi"] & \rho_S(t)
\end{tikzcd}
$$

Geometric transformation of Bloch Vector under a Quantum Map

Since $\Phi : \rho \longrightarrow \rho'$ and by Bloch theorem any ρ can be represented by $\vec{v}$

$$\implies \Phi : \vec{v} \longrightarrow \overrightarrow{v'} = M\vec{v} + c$$

$$M_{ij} = \frac{1}{2} \sum_{\alpha} \mathrm{Tr}(\sigma_i K_\alpha \sigma_j K_\alpha^\dagger)$$

$$c_i = \frac{1}{2} \sum_{\alpha} \mathrm{Tr}(\sigma_i K_\alpha K_\alpha^\dagger) \, .$$

The Phase Damping Map:

The phase damping map is:

$$\Phi(\rho') = p\rho + (1-p)Z\rho Z \ , \tag{179}$$

where $Z \equiv \sigma_z$, so the Kraus operators are $K_0 = \sqrt{p}I$ and $K_1 = \sqrt{1-p}Z$. This map can be understood as

$$\rho \mapsto \rho' = \begin{cases} \rho & \text{w/ prob. } p \\ Z\rho Z & \text{w/ prob. } 1-p \end{cases} \tag{180}$$

Using our general result, Eq. (174) we have in this case:

$$c_i = \frac{1}{2}\sum_\alpha \mathrm{Tr}(\sigma_i K_\alpha K_\alpha^\dagger) = \frac{1}{2}[p\mathrm{Tr}(\sigma_i) + (1-p)\mathrm{Tr}(\sigma_i)] = 0 \tag{181}$$

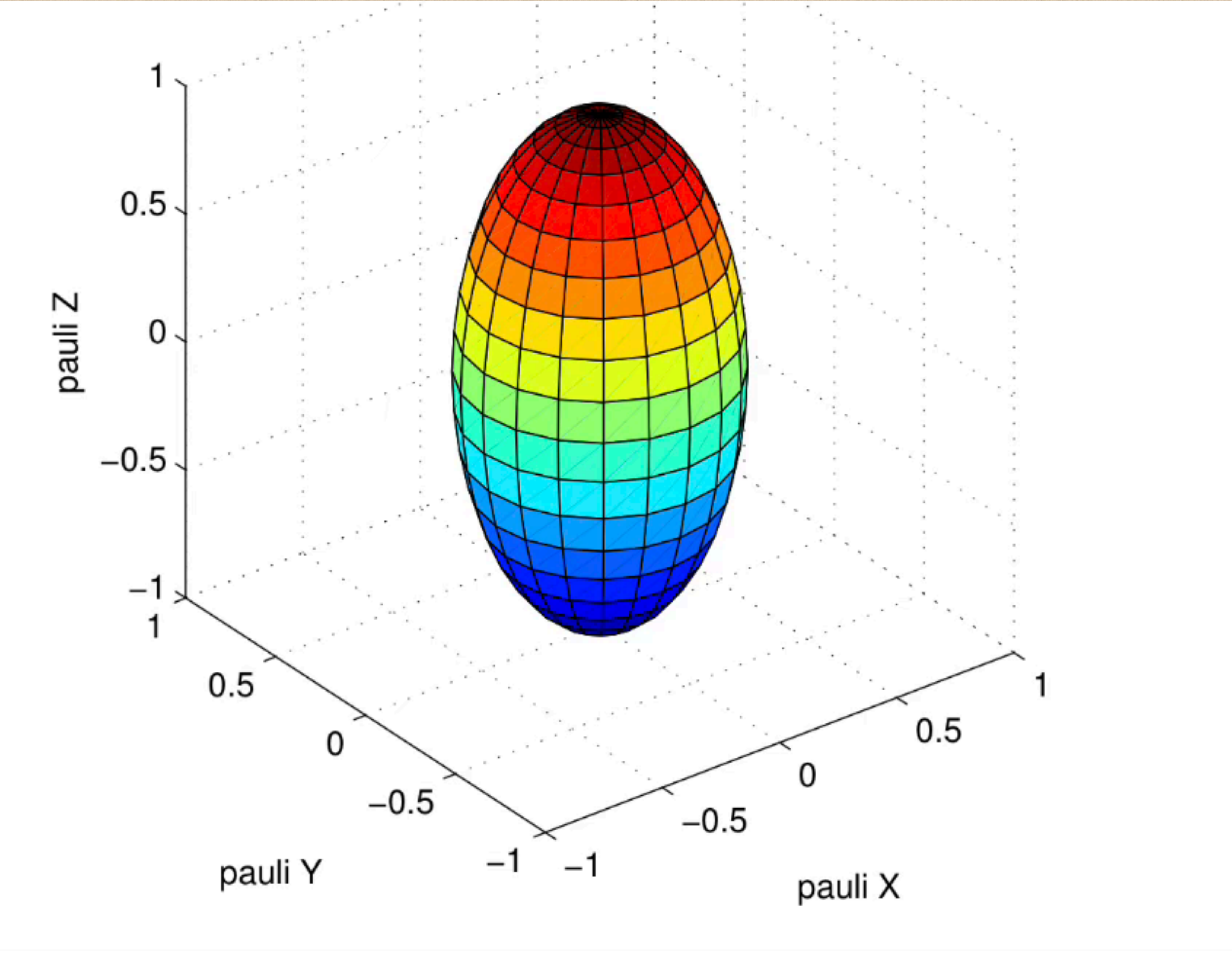
[in agreement with the fact that the phase damping map is unital; recall Eq. (178)], and:

$$M_{ij} = \frac{1}{2}\sum_\alpha \mathrm{Tr}(\sigma_i K_\alpha \sigma_j K_\alpha^\dagger) = \frac{1}{2}[p\mathrm{Tr}(\sigma_i\sigma_j) + (1-p)\mathrm{Tr}(\sigma_i Z\sigma_j Z)] = p\delta_{ij} + \frac{1}{2}(1-p)J_{ij} \ , \tag{182}$$

$$M = \mathrm{diag}[p-(1-p), p-(1-p), p+(1-p)] = \begin{pmatrix} 2p-1 & 0 & 0 \\ 0 & 2p-1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (184)$$

and

$$\vec{v}' = M\vec{v} = [(2p-1)v_x, (2p-1)v_y, v_z]^t \ . \quad (185)$$



ρ for qubit $|\psi\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle + |\downarrow\rangle\right)$

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\sigma_z \rho \sigma_z = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\sigma_z \rho \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1/2 & -1/2 \\ 1/2 & -1/2 \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$$

$$\rho \longrightarrow \rho' = p\rho + (1-p)\,\sigma_z \rho \sigma_z = \frac{1}{2}\begin{bmatrix} 1 & (2p-1) \\ (2p-1) & 1 \end{bmatrix}$$

Afte N ϕ mapping:

$$\rho^{N} = \frac{1}{2}\begin{bmatrix} 1 & (2p-1)^N \\ (2p-1)^N & 1 \end{bmatrix}$$ if each scattering toook δt & $p = -\Gamma \delta t$

then $(2p-1)^N = \left(\frac{2t\Gamma}{N} - 1\right)^N = (-1)^N e^{-2\Gamma t}$

$$\boxed{\rho(t) = \frac{1}{2}\begin{bmatrix} 1 & (-1)^N e^{-\Gamma t} \\ (-1)^N e^{-\Gamma t} & 1 \end{bmatrix}}$$

Bit-Flipping QM:

D. The Bit Flip Map

The bit flip map is:

$$\rho \mapsto \rho' = \begin{cases} \rho & \text{w/ prob. } p \\ X\rho X & \text{w/ prob. } 1-p \end{cases}$$

$$\vec{v} \mapsto \vec{v}\,' = [v_x, (2p-1)v_y, (2p-1)v_z] = M\vec{v} + \vec{c}\ , \tag{191}$$

where

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2p-1 & 0 \\ 0 & 0 & 2p-1 \end{pmatrix} \tag{192a}$$

$$\vec{c} = \bar{0}\ . \tag{192b}$$

Depolarizing QM:

$$\rho \mapsto \rho' = \begin{cases} \frac{1}{2}I & \text{w/ prob. } p \\ \rho & \text{w/ prob. } 1-p \end{cases} . \tag{193}$$

$$M = (1-p)I$$
$$\vec{c} = \bar{0} .$$

Amplitude Damping/Spontaneous emission

$$|0\rangle \mapsto |0\rangle \quad \text{with probability } 1 \tag{202a}$$

$$|1\rangle \mapsto |0\rangle \quad \text{with probability } p \tag{202b}$$

$$M = \begin{pmatrix} \sqrt{1-p} & 0 & 0 \\ 0 & \sqrt{1-p} & 0 \\ 0 & 0 & 1-p \end{pmatrix} .$$

$$\vec{c} = (0, 0, p) .$$

Lindblad Master Equation:

Just like how Schrödinger equation tells us about the evolution of the system

Lindblad Master Equation tells us how the Open System evolve.

A. Derivation

By Taylor expansion around $t = 0$ we have:

$$\rho(dt) = \rho(0) + \dot{\rho}|_0 dt + O(dt^2) \ . \tag{254}$$

On the other hand, the Kraus OSR tells us that:

$$\rho(dt) = \sum_\alpha K_\alpha(dt)\rho(0)K_\alpha^\dagger(dt) \ . \tag{255}$$

one of the Kraus operators must contain the identity operator. Thus, let us write

$$K_0 = I + L_0 dt \ , \tag{256}$$

so that

$$K_0 \rho(0) K_0^\dagger = \rho(0) + [L_0 \rho(0) + \rho(0) L_0^\dagger] dt + O(dt^2) \ . \tag{257}$$

This contributes one term of order dt, but there must be more (since as we know a Kraus OSR with a single Kraus operator is equivalent to unitary evolution). Thus, we can pick all other Kraus operators as

$$K_\alpha = \sqrt{dt} L_\alpha \ , \quad \alpha \geq 1 \ , \tag{258}$$

so that

$$K_\alpha \rho(0) K_\alpha^\dagger = L_\alpha \rho(0) L_\alpha^\dagger dt \ . \tag{259}$$

Let us now enforce the normalization condition $\sum_{\alpha=0} K_\alpha^\dagger K_\alpha = I$, up to $O(dt)$:

$$I = K_0^\dagger K_0 + \sum_{\alpha \geq 1} K_\alpha^\dagger K_\alpha = I + dt \left(L_0 + L_0^\dagger + \sum_{\alpha \geq 1} L_\alpha^\dagger L_\alpha \right) + O(dt^2) \ . \tag{260}$$

$$I = K_0^\dagger K_0 + \sum_{\alpha\geq 1} K_\alpha^\dagger K_\alpha = I + dt\left(L_0 + L_0^\dagger + \sum_{\alpha\geq 1} L_\alpha^\dagger L_\alpha\right) + O(dt^2)\ . \tag{260}$$

Without loss of generality we can decompose the general operator L_0 into a Hermitian and anti-Hermitian part: $L_0 = A - iH$, with $A = A^\dagger$ and $H = H^\dagger$. Thus, Eq. (260) tells us that to $O(dt)$:

$$A = -\frac{1}{2}\sum_{\alpha\geq 1} L_\alpha^\dagger L_\alpha\ . \tag{261}$$

Plugging all this back into the Kraus OSR, Eq. (255), we find:

$$\begin{aligned}
\rho(dt) &= K_0\rho(0)K_0^\dagger + \sum_{\alpha\geq 1} K_\alpha\rho(0)K_\alpha^\dagger && \text{(262a)}\\
&= \rho(0) + (A - iH)dt\rho(0) + \rho(0)(A + iH)dt + \sum_{\alpha\geq 1} L_\alpha\rho(0)L_\alpha^\dagger dt + O(dt^2) && \text{(262b)}\\
&= \rho(0) - i[H,\rho(0)]dt + \{A,\rho(0)\}dt + \sum_{\alpha\geq 1} L_\alpha\rho(0)L_\alpha^\dagger dt + O(dt^2) && \text{(262c)}\\
&= \rho(0) - i[H,\rho(0)]dt + \sum_{\alpha\geq 1}\left(L_\alpha\rho(0)L_\alpha^\dagger - \frac{1}{2}\left\{L_\alpha^\dagger L_\alpha,\rho(0)\right\}\right)dt + O(dt^2)\ . && \text{(262d)}
\end{aligned}$$

Therefore:

$$\dot\rho(t)|_0 = \lim_{dt\to 0}\frac{\rho(dt)-\rho(0)}{dt} = -i[H,\rho(0)] + \sum_{\alpha\geq 1}\left(L_\alpha\rho(0)L_\alpha^\dagger - \frac{1}{2}\left\{L_\alpha^\dagger L_\alpha,\rho(0)\right\}\right)\ . \tag{263}$$

$$\dot{\rho}(t)|_0 = -i[H, \rho(0)] + \sum_{\alpha \geq 1} \gamma_\alpha \left(L_\alpha \rho(0) L_\alpha^\dagger - \frac{1}{2} \left\{ L_\alpha^\dagger L_\alpha, \rho(0) \right\} \right) . \tag{264}$$

This result is valid as a short time expansion near $t = 0$. We now make an extra, very significant assumption:

Assumption 1. *Eq.* (264) *is valid for* all *times* $t > 0$.

This is essentially the Markovian limit, which states (informally) that there is no memory in the evolution, as manifested by the fact that the evolution "resets" every dt. It is motivated in part by the observation that if we limit our attention just to $\dot{\rho}(t)|_0 = -i[H, \rho(0)]$, then we already know that this replacement is valid, i.e., that we can indeed replace this with $\dot{\rho}(t) = -i[H, \rho(t)]$ for all t, since this is just the Schrödinger equation. With this we finally arrive at the *Lindblad equation*:

$$\frac{d\rho}{dt} = -i[H, \rho(t)] + \sum_{\alpha} \gamma_\alpha \left(L_\alpha \rho(t) L_\alpha^\dagger - \frac{1}{2} \left\{ L_\alpha^\dagger L_\alpha, \rho(t) \right\} \right) \equiv \mathcal{L}\rho . \tag{265}$$

Note : this part below , gives us Non-Unitary Evolution called Dissipator

$$\mathcal{L}_D[\cdot] = \sum_{\alpha} \gamma_\alpha \left(L_\alpha \cdot L_\alpha^\dagger - \frac{1}{2} \{ L_\alpha^\dagger L_\alpha, \cdot \} \right) , \quad \gamma_\alpha \geq 0 . \tag{266}$$

B. The Markovian evolution operator as a one-parameter semigroup

The formal solution of the (Lindblad) equation $\dot{\rho}(t) = \mathcal{L}\rho$ is

$$\rho(t) = e^{\mathcal{L}t}\rho(0) \equiv \Lambda_t \rho(0) , \tag{268}$$

where Λ_t is called the Markovian evolution operator (it is also a quantum map). The set $\{\Lambda_t\}_{t\geq 0}$ forms a one-parameter semigroup. The one-parameter part is clear: the set depends only on the time t, once the Lindblad generator $\mathcal{L}$ is fixed. The reason this is a semi-group is that the superoperators Λ_t only satisfy three of the four properties of a group:

1. Identity operator: $\Lambda_0 = \mathcal{I}$.
2. Closed under multiplication: $\Lambda_t \Lambda_s = e^{\mathcal{L}t} e^{\mathcal{L}s} = e^{\mathcal{L}(t+s)} = \Lambda_{t+s}$.
3. Associative: $(\Lambda_t \Lambda_s)\Lambda_r = \Lambda_t(\Lambda_s \Lambda_r)$.

However, not every element has an inverse: as we shall see, complete positivity forces all the eigenvalues of $\mathcal{L}$ to be non-positive, so that the map Λ_t is contractive, corresponding to exponential decay. This means that Λ_∞ has at least one zero eigenvalue, so it does not possess an inverse. We shall shortly see this in examples.

We already encountered the phase damping model in the Kraus OSR setting in Sec. VII C. Let us now study a Lindblad equation model that generates the same map.

Let $L_1 = \sigma^z = Z$, $\gamma_1 = \gamma$, $\gamma_{\alpha\geq 2} = 0$, and $H = 0$. Thus,

$$\dot{\rho}(t) = \gamma(Z\rho Z^\dagger - \frac{1}{2}\{Z^\dagger Z, \rho\}) = \gamma(Z\rho Z - \rho) \ . \tag{277}$$

Using $\rho = \frac{1}{2}(I + \vec{v}\cdot\vec{\sigma})$, the left-hand side evaluates to $\dot{\rho} = \frac{1}{2}\dot{\vec{v}}\cdot\vec{\sigma}$. For the right hand side $Z\rho Z = \frac{1}{2}(I - v_x X - v_y Y + v_z Z)$, and we thus arrive at:

$$\frac{1}{2}\left(\dot{v}_x X + \dot{v}_y Y + \dot{v}_z Z\right) = -\gamma\left(v_x X + v_y Y\right) \ . \tag{278}$$

Equating the two sides componentwise (multiply both sides by X, Y, or Z, and take the trace) gives:

$$\dot{v}_x = -2\gamma v_x \implies v_x(t) = v_x(0)e^{-2\gamma t} \tag{279a}$$

$$\dot{v}_y = -2\gamma v_y \implies v_y(t) = v_y(0)e^{-2\gamma t} \tag{279b}$$

$$\dot{v}_z = 0 \implies v_z(t) = v_z(0) \ . \tag{279c}$$

We can see that, since $\gamma \geq 0$, the map is contractive, and the Bloch sphere collapses to the v_z-axis exponentially fast with time. In the limit $t \to \infty$, this simply projects every state directly to the v_z-axis, which is manifestly uninvertible.

We can now match the Lindblad equation solution to the Kraus OSR result from Sec. VII C, where we found the Kraus operators $K_0 = \sqrt{p}I$ and $K_1 = \sqrt{1-p}Z$, and found that the Bloch vector is mapped to

$$\vec{v}' = ((2p-1)v_x(0), (2p-1)v_y(0), v_z(0)) \equiv (v_x(t), v_y(t), v_z(t)) \ . \tag{280}$$

The Lindblad phase damping result and the Kraus OSR thus have exactly the same effect provided we identify

$$2p - 1 = e^{-2\gamma t} \implies p(t) = \frac{1}{2}(1 + e^{-2\gamma t}) \ . \tag{281}$$

The probability in this model approaches 1/2 in the limit $t \to \infty$.

If we now allow $H \neq 0$, i.e., solve the full Lindblad equation $\dot{\rho} = -i[H,\rho] + \gamma(Z\rho Z^\dagger - \rho)$, then the result in Sec. IX D 1 shows that this gives rise to a rotating Bloch ellipsoid that is simultaneously shrinking exponentially along its principal axis.